

~~Let~~

Directed Graphs / Digraphs.

Let ~~to be~~ V be a finite non empty ~~a directed~~ ^{set} ~~graph~~ ^{digraph or directed graph} G ^{on V}
is made up of elements of V (called the vertices or nodes of G). A subset E of $V \times V$ contains the directed edges or arcs of G . The set V is called vertex set of G and set E is called the ^{set} edges of " G ".
 $G = (V, E)$ denotes a directed graph.

Properties of Digraph

- ① Let R be a relation from set A to set B . Defined as $(a, b) \in R$ then in a digraph it is represented as an edge $A \rightarrow B$.
- ① Relation R is reflexive if there is a loop at every node of the directed graph.
- ② Relation R is irreflexive if there is no loop at any ~~a~~ node of directed graph.
- ③ Relation R is symmetric if for every edge b/w distinct nodes, an edge is always present in the opposite direction.
- ④ A Relation R is anti symmetric if there are never two edges in the opposite direction b/w the distinct nodes.
- ⑤ A Relation R is transitive, if there is an edge from A to B and B to C then there is always an edge b/w $A \rightarrow C$.

P-1) Representing ~~vertex~~ vertex as point.

P-2) Delete all the self Loops. $(4,4), (5,5), (6,6), (7,7)$.

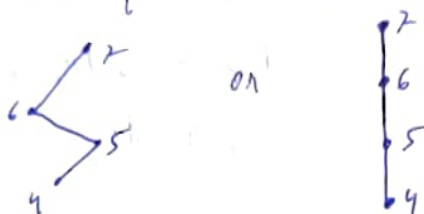
P-3) Removal of transitive edges.
 $(4,7), (4,6), (5,7)$.

P-4) Eliminate the arrows.

→ necessary to write Hasse

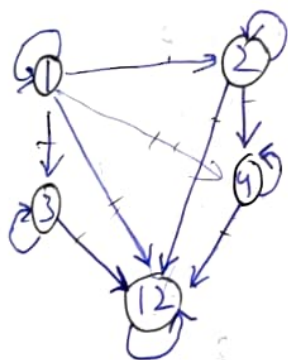
$(4 \rightarrow 5, 5 \rightarrow 6), 4 \rightarrow 6$
 $(4 \rightarrow 6, 6 \rightarrow 7), (4 \rightarrow 7)$
 $(5 \rightarrow 6, 6 \rightarrow 7), (5 \rightarrow 7)$.

Hasse Relation $\Rightarrow \{(4,5), (5,6), (6,7)\}$.



Q SET $A = \{1, 2, 3, 4, 12\}$. Generate Relation R such that a is related to b if A divides B . Draw the Hasse diagram for same.

$R = \{(1,1), (1,2), (1,3), (1,4), (1,12), (2,2), (2,4), (2,12), (3,3), (3,12), (4,4), (4,12), (12,12)\}$.



P-1) Representing vertex as points.

P-2) Delete all the self Loops

$(1,1), (2,2), (3,3), (4,4), (5,5)$

P-3) Removal of transitive edges.

$(1,2), (2,4) \rightarrow (1,4)$

$(1 \rightarrow 3), (3,12) \rightarrow (1,12)$

$(1 \rightarrow 4), (4,12) \rightarrow (1,12)$

$(2,3), (3,12) \rightarrow (2,12)$

$(1,4), (1,12), (2,12)$

P-4) Eliminate the arrows.

Hasse Relation $\Rightarrow \{(1,2), (2,4), (3,12), (4,12)\}$



Hasse Diagram.

It is a graphical representation of a relation on the elements of a partially ordered set [POSET] with ~~arrow~~ implied upward orientation. A point is drawn for each element of the POSET and joined with the line segment according to the following rules.

- ① If (element $p < \text{element } q$) ($p < q$) in the POSET then the point corresponding to p appears lower in the drawing.
- ② The two points p and q will be joined by a line segment, if p is related to q ($p R q$).

Conversion of Digraph to Hasse Diagram

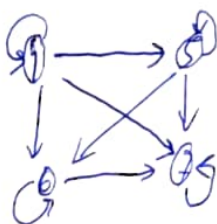
- ① The vertices in the Hasse diagram are denoted by points rather than this circle.
- Since a partial order relation is reflexive each vertex of set A must be related to itself so the edges from the vertex to itself are deleted in the Hasse diagram.
- ② Since a partial order relation is transitive whenever a is related to b ($a R b$) b is related to c ($b R c$) we have a is related to c ($a R c$).
- ③ eliminate all the edges that are implied by the transitive property in Hasse, that is delete the edge A to C , and retain A to B & B to C .

If a vertex a is connected to vertex b by an edge, that is (a is related to b) ($a R b$), that vertex a appears above b .

The arrows will be omitted from the edges in the Hasse diagram.

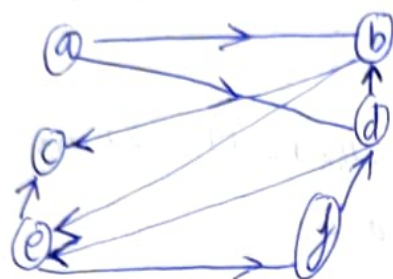
Construct a diagram for $R = \{ (4, 5), (4, 6), (4, 7), (5, 6), (5, 7), (6, 7), (4, 4), (5, 5), (6, 6), (7, 7) \}$ on a set $A = \{ 4, 5, 6, 7 \}$.

to convert the obtained digraph into Hasse Diagram.

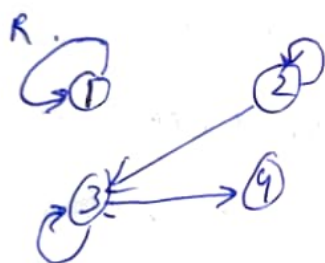


Digraph

Q Draw the digraph for $G_1 = (V_1, E_1)$ where $V_1 = \{a, b, c, d, e, f\}$ and $E_1 = \{(a, b), (a, d), (b, c), (b, e), (d, b), (d, c), (d, e), (e, c), (e, f), (f, d)\}$.



for set $A = \{1, 2, 3, 4\}$, let relation $R = \{(1, 1), (2, 2), (2, 3), (3, 3), (3, 4)\}$, draw the digraph with graph G that is associated with R, R^2, R^3, R^4 .



ROR: $\{(1, 1), (2, 2), (2, 3), (3, 3), (3, 4)\}$

$$R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R^2 = \{(1, 1), (2, 2), (3, 3), (4, 3), (3, 4)\}$$

Q draw the Hasse diagram: D_{12} & D_{45} with Relation A Divides B

Positive divisors of 12

Positive divisors of 45

$$X \Rightarrow \{1, 2, 3, 4, 6, 12\} \quad | \quad Y \Rightarrow \{1, 3, 5, 9, 15, 45\}$$

$$R \Rightarrow \{(1,1), (1,2), (1,3), (1,4), (1,6), (1,12), (2,4), (2,6), (2,12), (3,6), (3,12), (4,12), (6,12)\}$$

P1) Representing vertex as points.

P2) Removing all the self loops.

$$(1,1), (2,2), (3,3), (4,4), (6,6), (12,12)$$

P3) Removing all the transitive edge.

$$(1,2), (2,4) \rightarrow (1,4)$$

$$(1,4), (2,6) \rightarrow (1,6)$$

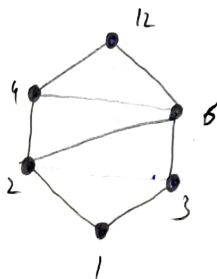
$$(1,6), (6,12) \rightarrow (1,12)$$

$$(2,4), (4,12) \rightarrow (2,12)$$

$$(3,6), (6,12) \rightarrow (3,12)$$

P4) ~~Removing~~ Eliminating all arrows.

$$R \Rightarrow \{(1,2), (1,3), (2,4), (2,6), (3,6), (4,12), (6,12)\}$$



$$R = \{(1,1), (3,3), (5,5), (9,9), (15,15), (45,45), (1,3), (1,5), (1,9), (1,15), (1,45), (3,9), (3,15), (3,45), (5,15), (5,45), (9,45), (15,45)\}$$

P1) Representing vertex as points.

P2) Removing all the self loops

$$(1,1), (3,3), (5,5), (9,9), (15,15), (45,45)$$

P3) Removing all the transitive edge.

$$(1,3), (3,9) \rightarrow (1,9)$$

$$(1,5), (5,15) \rightarrow (1,15)$$

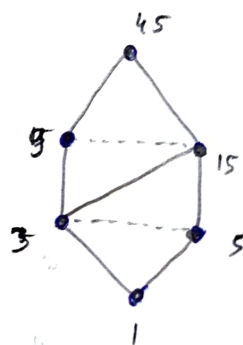
$$(1,15), (15,45) \rightarrow (1,45)$$

$$(3,9), (9,45) \rightarrow (3,45)$$

$$(5,15), (15,45) \rightarrow (5,45)$$

P4) Eliminating all arrows.

$$R \Rightarrow \{(1,3), (1,5), (3,9), (3,15), (5,15), (9,45), (15,45)\}$$



Q Draw the Hasse diagram positive divisors of 36 & the relation is a divides b

$$A = \{1, 2, 3, 4, 6, 9, 12, 36, 18\}, P$$

$$R \Rightarrow \{(1, 1), (2, 2), (3, 3), (4, 4), (6, 6), (9, 9), (12, 12), (36, 36), (18, 18)$$

$$(1, 2), (1, 3), (1, 4), (1, 6), (1, 9), (1, 12), (1, 36), (1, 18)$$

$$(2, 4), (2, 6), (2, 12), (2, 36), (2, 18)$$

$$(3, 6), (3, 9), (3, 12), (3, 36), (3, 18)$$

$$(4, 12), (4, 36)$$

$$(6, 12), (6, 36), (9, 36), (12, 36), (6, 18), (9, 18), (18, 36)$$

P_1 Representing vertex as points.

P_2 Remove all the self loops.

$$\{(1, 1), (2, 2), (3, 3), (4, 4), (6, 6), (9, 9), (12, 12), (36, 36)\}$$

P_3 Removal of transitive ~~edges~~ edges.

$$(1, 2), (2, 4) \rightarrow (1, 4)$$

$$(1, 2), (2, 6) \rightarrow (1, 6)$$

$$(1, 12), (2, 12) \rightarrow (1, 12)$$

$$(1, 2), (2, 36) \rightarrow (1, 36)$$

$$(2, 4), (4, 12) \rightarrow (2, 12)$$

$$(2, 4), (4, 36) \rightarrow (2, 36)$$

$$(3, 6), (6, 12) \rightarrow (3, 12)$$

$$(3, 6), (6, 36) \rightarrow (3, 36)$$

$$(9, 12), (12, 36) \rightarrow (9, 36)$$

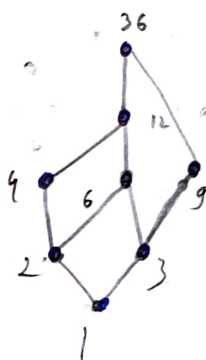
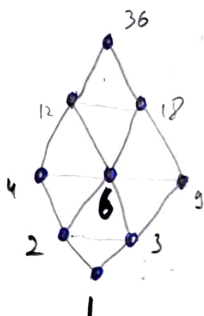
$$(6, 12), (12, 36) \rightarrow (6, 36)$$

P_4 Eliminate all the arrows.

$$\text{final } R \Rightarrow \{(1, 2), (1, 3), (2, 4), (2, 6), (3, 6), (3, 9),$$

$$(4, 12), (6, 12), (9, 36), (12, 36)\}$$

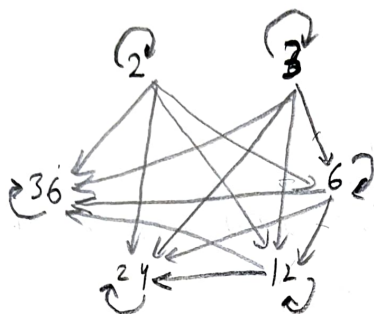
$$(6, 18), (9, 18), (18, 36)\}$$



A set x be $\{2, 3, 6, 12, 24, 36\}$ $A \mid_R B$ if A divides B
 draw the digraph and convert the obtained digraph to Hasse diagram

$x = \{2, 3, 6, 12, 24, 36\}$, ARB a divides B .

$R = \{(2, 6), (2, 12), (2, 24), (2, 36), (3, 6), (3, 12), (3, 24), (3, 36), (6, 12), (6, 24), (6, 36), (12, 24), (12, 36), (24, 36), (2, 3), (3, 2), (6, 6), (12, 12), (24, 24), (36, 36)\}$



P1) Representing vertex as points

P2) Delete all the self loops.

$(2, 2), (3, 3), (6, 6), (12, 12), (24, 24), (36, 36)$

P3) Removal of transitive edge.

$(3, 6), (6, 12) \rightarrow (3, 12)$

$(6, 12), (12, 24) \rightarrow (6, 24)$

$(6, 12), (12, 36) \rightarrow (6, 36)$

$(3, 12), (12, 24) \rightarrow (3, 24)$

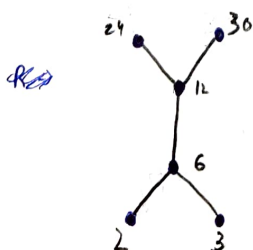
$(2, 6), (6, 12) \rightarrow (2, 12)$

$(2, 12), (12, 24) \rightarrow (2, 24)$

$(2, 6), (6, 36) \rightarrow (2, 36)$

$(2, 12), (12, 36) \rightarrow (2, 36)$

$(3, 12), (12, 36) \rightarrow (3, 36)$



P4) Eliminate the arrows.

$R = \{(2, 6), (3, 6), (6, 12), (12, 24), (12, 36)\}$